EXPERIMENT 2
MONOALPHABETIC SUBSTITUTION CIPHER

AIM

To implement the Monoalphabetic Substitution Cipher using C.

ALGORITHM

1. Read the plaintext from the user.
2. Define a substitution key containing all 26 unique alphabet letters.
3. Replace each plaintext letter with its corresponding letter from the substitution key.
4. Display the encrypted (cipher) text.
5. For decryption, replace each cipher letter with its corresponding original alphabet letter.
6. Display the decrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char plain[100];
    char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
    int i;

    printf("Enter Plain Text: ");
    scanf("%s", plain);

    printf("Encrypted Text: ");
    for(i=0; plain[i]!='\0'; i++)
        printf("%c", key[plain[i]-'A']);

    printf("\nDecrypted Text: ");
    for(i=0; plain[i]!='\0'; i++) {
        int j;
        for(j=0;j<26;j++)
            if(key[j]==key[plain[i]-'A'])
                break;
        printf("%c",'A'+j);
    }

    return 0;
}
```

OUTPUT

```
Output                                                    Clear
Enter Plain Text: GOOD MORNING
Encrypted Text: UGGR
Decrypted Text: GOOD

=== Code Execution Successful ===
```

RESULT

Thus, the Monoalphabetic Substitution Cipher was implemented successfully using C.